

DETAILED DESCRIPTION OF THE INVENTION — MULTI-NODE SYSTEM COORDINATION AND SYNCHRONIZED PHYSIOLOGIC ANALYSIS

The intraluminal physiologic monitoring device may operate as one node within a coordinated multi-node physiologic monitoring system configured to analyze relationships between signals collected from multiple body regions or multiple devices. System coordination may enable synchronized physiologic analysis extending beyond isolated measurement.

In various embodiments, the device may communicate with additional sensing nodes positioned on external body regions, wearable devices, handheld scanners, or companion intraluminal devices. Multi-node systems may include penile monitoring devices, chest or cardiovascular monitoring nodes, motion tracking devices, or environmental sensors contributing contextual information.

Synchronization mechanisms may align timestamps, sampling intervals, or event markers across devices to permit correlation of physiologic responses occurring simultaneously or sequentially. Coordinated analysis may reveal system-level physiologic interactions not observable through single-node monitoring.

Data fusion frameworks may combine measurements from multiple nodes to generate composite indices representing interaction dynamics, response timing relationships, or systemic physiologic patterns. Multi-node fusion may enhance interpretation accuracy and predictive capability.

Certain embodiments may designate one device as a coordination hub responsible for aggregating signals from secondary nodes. Alternatively, distributed coordination architectures may allow peer-to-peer synchronization without reliance on a central controller.

Communication between nodes may occur through wireless links, relay devices, companion applications, or cloud-based synchronization services. Latency compensation algorithms may correct timing differences between devices.

Multi-node analysis systems may support compatibility modeling, performance evaluation, therapeutic monitoring, or research applications involving correlated physiologic signals across individuals or body systems.

Adaptive coordination algorithms may learn optimal synchronization strategies over time based on device placement, user behavior, or signal characteristics. Continuous learning may improve system coherence during repeated use.

The multi-node coordination and synchronized physiologic analysis concepts described herein are illustrative rather than limiting. Any system enabling coordinated analysis of physiologic signals across multiple sensing devices is considered within the scope of the present disclosure.